Xuechao ZHANG

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Summary

Robotics and AI researcher with a background bridging hardware and software, specializing in dexterous manipulation. Proven record from research to production — with industry experience (**Apple, Bosch, Tencent**) and publications at top conferences (**CoRL, IROS**).

Education

- **Arizona State University** — Jan. 2025 - Dec. 2026 (Expected)
  Ph.D. Student (Transitioned to Master), Robotics Engineering ***GPA:*** *4.0/4.0* — *Arizona, USA*
- **Shanghai Jiao Tong University** — Sept. 2021 - Mar. 2024
  Master, Electronic Information ***GPA:*** *3.6/4.0* — *Shanghai, China*
- **Georgia Institute of Technology** — Jul. 2018 - Aug. 2018
  Exchange Student in Atlanta Summer Program ***GPA:*** *3.9/4.0* — *Atlanta, USA*
- **Southeast University** — Sept. 2017 - Jun. 2021
  Bachelor, Robotics Engineering (Chien-Shiung Wu College) ***GPA:****3.7/4.0* — *Nanjing, China*

Work Experience

- **Sim-to-Real Reinforcement Learning for Robotic Interaction** — May. 2025 - Present
  Applied AI Software Engineer Intern @ Apple Inc. — *Cupertino, USA*
  - Contributed to an Embodied AI initiative focused on autonomous robotic manipulation for dynamic device interaction, with emphasis on reliable policy learning and physical deployment readiness.
  - Engineered massively parallel simulation pipelines using Isaac Sim to train reinforcement learning policies for **multi-fingered dexterous hands** and robotic arms for arbitrary-pose object grasping.
  - Integrated multimodal visual and tactile sensing, and applied **domain randomization** techniques to improve sim-to-real transfer robustness for physical robotic deployment.
  - Collaborated on data collection pipelines for fine-tuning open-source **Vision-Language-Action (VLA) models**, and benchmarked their performance against RL-based policies.

- **Silicon Validation Automation System** — Apr. 2024 - Jan. 2025
  Robotics Engineering Intern @ Apple Inc. — *Shanghai, China*
  - Architected and delivered a **UR10e**-based automation system for USB interoperability validation. Task included device ports localization, cable plugging (e.g., Type-C, HDMI) and simulating user operations on HID devices.
  - Implemented and fine-tuned state-of-the-art **6-DoF visual localization**, achieving precision of **0.3 mm/2°** with marker and **2.0 mm/5°** without markers at 1 m working distance.
  - Fused camera and force/torque feedback for contact-aware control, reaching **99.2%** grasp success (universal boxes) and **96.5%** cable-insertion success over **200+** device SKUs in a full regression pass.
  - Developed a **Python**-based robotics toolkit covering system calibration, front-end task management, robot path planning and visualization, plus robust fault-recovery and logging, enabling **24/7** reliable operation.
  - Deployed across labs in **Shanghai/Tokyo/Cupertino**; results presented to **VP**-level leadership.

- **Autonomous Driving Car Demonstration Model** — Aug. 2023 - Jan. 2024
  Strategic Intern (Part-time) @ Bosch (China) Investment Ltd. — *Shanghai, China*
  - Designed and implemented an autonomous indoor navigation platform using a Raspberry Pi-powered demonstration vehicle, equipped with onboard camera for real-time perception and control.
  - Developed educational materials and documentation as part of the global Bosch AI Learning Curriculum, and conducted online and offline training sessions for internal staff, reaching over **100** employees.

- **Swarm Robots System for Cooperative Construction** — Jun. 2022 - Sept. 2022
  Research Intern @ Tencent Robotics X Lab — *Shenzhen, China*
  - Contributed to the development of a digital-twin system for heterogeneous robots, including **quadruped robots and Mecanum wheeled robots**, which can transport blocks/slopes to construct multi-layer buildings.
  - Designed the mechanical and electrical systems and developed the embedded software based on the Robot Operating System (ROS) to enable Mecanum wheeled robots to operate blocks/slopes under centralized control.
  - Integrated the visual sensors and IMU of the robots using Kalman filtering to enable autonomous localization, reducing positioning error by **43%** and improving positioning information reporting rate by **57%**.

Research Experience

- **Generative Dexterous Manipulation via Context-Aware Policies** Aug. 2025 - Present
  Research Assistant @ Interactive Robotics Laboratory *Advised by Prof. Heni Ben Amor*
  - Developed the lab's **dexterous research infrastructure** by evaluating and acquiring state-of-the-art robotic hands and data-collection gloves. Implemented real-time **kinematic retargeting** to map human dexterity onto multi-fingered robotic hands for large-scale data collection.
  - Exploring generative policy based on **Transformer-based Conditional Variational Autoencoders (CVAE)** to model complex, multi-modal action distributions, enabling the robot to handle diverse object interactions with high precision.
  - Investigating **In-Context Learning (ICL)** mechanisms to adapt the policy's manipulation style online, utilizing historical interaction contexts to modulate behavior without explicit retraining (Work in Progress).

- **Model-Based Contact-Rich Dexterous Manipulation** Jan. 2025 - Jul. 2025
  Research Assistant @ Intelligent Robotics & Interactive Systems Lab *Advised by Prof. Wanxin Jin*
  - Conducted research on model-based planning for contact-rich dexterous manipulation, including fingertip in-air manipulation, and **LEAP Hand** on-palm reorientation.
  - Designed a real-time contact-implicit **Model Predictive Control (MPC)** framework based on a recently proposed complementarity-free multi-contact dynamic model. Implemented in **Python** with **CasADi**, sustaining **50–100 Hz** control, matching or exceeding reinforcement-learning baselines on challenging dexterous tasks.
  - Introduced a global sampling-based contact-position replanning layer on top of MPC, yielding a **bi-level** controller that escapes local minima; achieved **90.1%** average success on continuous-rotation tasks over a **20 object** test set.
  - Co-developed a **20 Hz 6-DoF pose tracker** for unknown dynamic objects in contact-rich scenes by integrating the contact dynamics into a visual-tracking pipeline; geometry is supervised via **Gaussian Splatting**, while physical parameters (mass, inertia, friction) are **updated online** via sampling-based optimization. Summarized in paper[1].

- **Active Perception and Robotic Grasp Prediction** Sept. 2022 - Aug. 2023
  Research Intern @ Shanghai Artificial Intelligence Laboratory *Advised by Dr. Dong Wang*
  - Conducted research on **robotics grasping prediction** in cluttered desktop scenes, focusing on active perception under partial observability.
  - Proposed **ACE-NBV**, an affordance-driven next-best-view policy that actively selects next camera poses which guide the arm toward feasible grasps for target objects.
  - Formulated a **multi-task learning** scheme over a shared implicit neural representation to couple grasp affordance and 3D reconstruction; leveraged the paradigm of novel view imagery from **NeRF** to predict grasp affordances for previously unobserved views.
  - Demonstrated significant improvements over current state-of-the-art methods by consistently identifying more informative views, resulting in a comparable grasp success rate with **32.4%** fewer observations.
  - Published a paper[2] as the first author in **CoRL 2023**.

- **Real-Time Digital Twin Platform of Multi-Robots** Feb. 2021 - Mar. 2022
  Graduate Researcher @ Shanghai Jiao Tong University IWIN-FINS Lab *Advised by Prof. Jianping He*
  - Contributed to the development of a multi-robot testbed that exploits the ideas of digital-twin system.
  - Designed and implemented a distributed PTZ camera network and AprilTag visual positioning system which can achieve **120 Hz** tracking frequency, **10 ms** delay, and **0.5 mm** tracking error.
  - Proposed a multi-camera sensing quality model and an optimization strategy for camera network configuration based on this model, which improves the overall positioning performance of the platform.
  - Published a paper[3] as the first author in **IROS 2022**, and submitted an invention patent.

- **Competition Robots Software and Hardware Development** Apr. 2019 - Nov.2020
  Team Leader @ Southeast University Smart Car Team
  - Designed a Mecanum wheeled chess-playing robot, which utilizes IMU and cameras for self-positioning, an electromagnetic system for moving chess pieces, and algorithms for solving the Eight Queens problem and playing the Quoridor. Video
  - Designed an intelligent car which is powered by supercapacitors, capable of wireless charging at 30 W through self-made circuits, and utilized inductors to detect alternating currents for navigation. Video
  - Designed a self-balancing bicycle robot, which was manufactured using 3D printing and utilizes an IMU and a flywheel system for balancing. Figure
  - Won the **Second Prize** of the 14th National College Student "NXP Cup" Smart Car Competition and **First Prize** of the 10th Jiangsu Provincial College Student Robotics Competition. Submitted **two** invention patents.

Publications & Activities

1. Yang W, Xie Z, **Zhang X**, et al. TwinTrack: Bridging Vision and Contact Physics for Real-Time Tracking of Unknown Dynamic Objects[C]//2026 IEEE International Conference on Robotics and Automation (ICRA). To appear. Paper Page Video
2. **Zhang X**, Wang D, Han S, et al. Affordance-Driven Next-Best-View Planning for Robotic Grasping[C]//Conference on Robot Learning (CoRL). PMLR, 2023: 2849-2862. Paper Page Video
3. **Zhang X**, Ding X, Ren Y, et al. Toward Global Sensing Quality Maximization: A Configuration Optimization Scheme for Camera Networks[C]//2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2022: 13386-13391. Paper Video
4. Hou Y, Li J, Fang Z, **Zhang X**. An Initialization Method of Deep Q-network for Learning Acceleration of Robotic Grasp[C]//2020 IEEE International Conference on Networking, Sensing and Control (ICNSC). IEEE, 2020: 1–6. Paper

- **Invited Reviewer:** IROS, ICRA, NeurIPS, RA-L

Honors and Awards

- Ira A. Fulton Schools of Engineering Graduate Scholarship, ASU — Aug. 2025 & Jan. 2026
- Outstanding Graduate of Shanghai Jiao Tong University — Mar. 2024
- First Class Academic Scholarship from Shanghai Jiao Tong University — Sept. 2021 & Sept. 2022
- SMC Corporation Scholarship — Sept. 2022
- "Chien-Shiung Student" of Southeast University (Top 1%) — Jan. 2021
- Second Prize of the 14th National "NXP Cup" Smart Car Competition — Aug. 2019
- Second Prize of Zhengbao Education Scholarship — Jun. 2019

Skills

- **Programming Languages**: Python, MATLAB; Additional: C++, CUDA, JAX, Swift, Verilog
- **Theoretical Knowledge**: Control Theory, Optimization, Deep Learning, Computer Vision, Computer Graphics
- **Robotics Experience**:
  - **Algorithms**: Model Predictive Control, SLAM, NeRF, Gaussian Splatting, Diffusion, Reinforcement Learning, Imitation Learning, VLA
  - **Systems & Tooling**: PyTorch, Docker, ROS, simulation (MuJoCo, Isaac Lab, SAPIEN, Simulink, Webots)
  - **Hardware & Platforms**: PCB design (Altium Designer), CAD (AutoCAD, SolidWorks, Fusion 360), 3D printing; Franka Emika, Universal Robots, NVIDIA Jetson